RESILIA MZ

Presentation guide: what to say, where to point and how to answer the jury

Resilient and efficient solar mini-grid for Mabote, Inhambane (Mozambique)

This guide prepares the team for the 6-minute presentation (5 slides). It contains the time plan, the script for each slide, an optional live demo of the platform, the jury's most likely questions with short answers, and a sheet of numbers to memorise.

98.6 % of real days with power all day	**17 h → 0 h** without power on a heavy-rain day	**+56 %** investment	**100 %** renewable energy

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Eduardo Mondlane University · EMERGE project · EMERGE Final Hackathon, Maputo
Platform: resilia-72h.streamlit.app · Code: github.com/EzequiasMiocha/resilia-72h

1. Before you start

The core idea, the 6-minute plan and preparation on the day.

The idea in 20 seconds

In Mabote, villages without grid electricity depend on solar mini-grids. A system designed only for the average day is cheap, but on cloudy and rainy days it leaves the village in the dark. **RESILIA MZ simulates, hour by hour, how the system balances supply and demand over 24 hours, and chooses the cheapest system that does not fail on bad days.**

The message the jury should take home

“The cheap system works on good days. Ours works every day.”

The 6-minute plan

No.	Slide	Time	Ends at	Speaker	Message
1	Context	50 s	0:50	Laura	The sun changes; bad days leave the village in the dark.
2	Challenge and objective	60 s	1:50	Laura	What we were asked, what we do and how it works.
3	Days in a row	100 s	3:30	Suely	4 real days: sunny days store energy for the rain.
4	Resilient and efficient	90 s	5:00	Ezequias	Resilient on 98.6 % of real days.
5	Conclusion	60 s	6:00	Ezequias	100 % renewable, free, next step: test in a village.

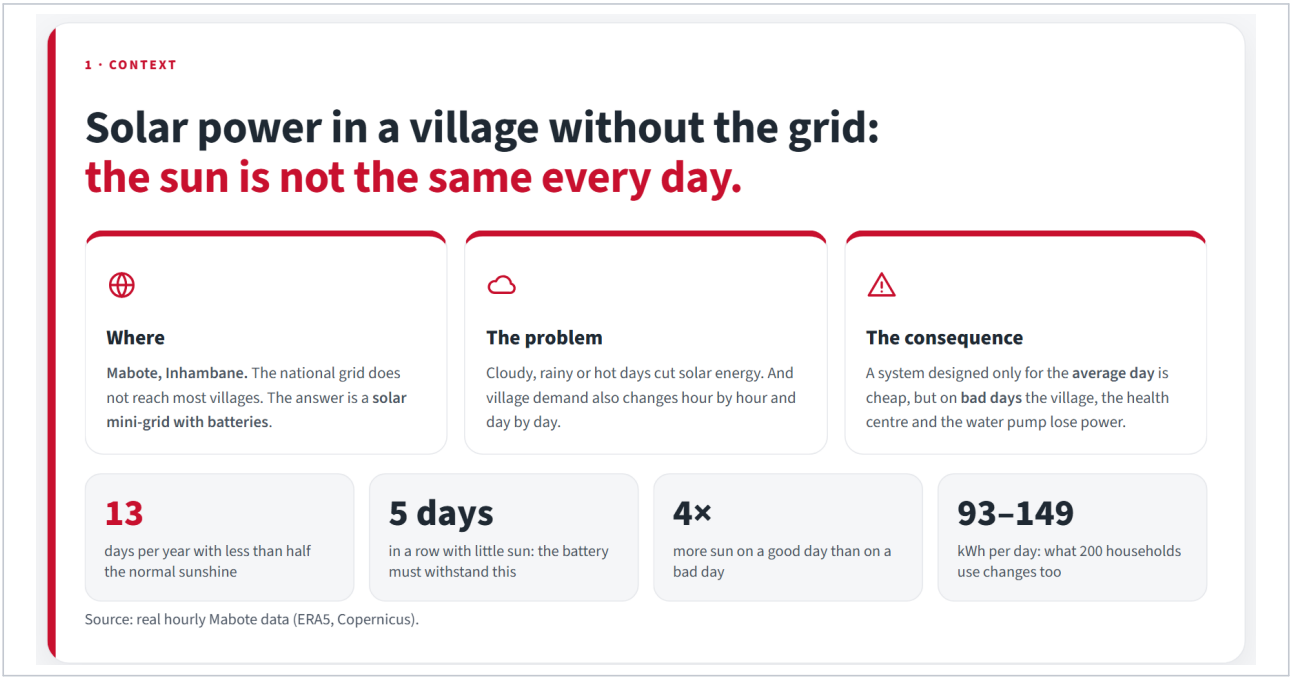
Order: **Laura** opens (1–2), **Suely** shows the days in a row (3) and **Ezequias** closes with the results (4–5). At each hand-over, the speaker who finishes says the hand-over sentence and the next one continues without a pause.

Preparation on the day

- Open the platform **before** entering the room, on the **Pitch** page, in the chosen language (PT / EN in the top right corner).
- Have a plan B: the platform running on the laptop (*streamlit run app.py*) and this PDF open.
- At the start, press **Start timer**: the clock shows when each slide should end and turns orange if you are late.
- The **Speaker notes** switch shows a short reminder under each slide.
- On slide 3, follow the **battery line** (at the bottom): it shows when the sun fills the battery and when it runs out.
- Rehearse at least twice with the timer. If time is short, shorten slide 2, never slide 3 or 4.

Slide 1 · Context

Time: 50 s · **Speaker:** Laura · **Goal:** the jury understands the problem.



What to say

Point to: the title

1. "In Mabote, in Inhambane province, the electricity grid does not reach most villages. The answer is a small grid of solar panels with batteries."

Point to: the card "The problem"

2. "But the sun is not the same every day. With clouds, rain or great heat there is less energy. And what the village uses also changes during the day."

Point to: the four numbers

3. "In Mabote there are about 13 days a year with less than half the normal sunshine, and up to 5 weak days can come in a row. On a good day there is almost four times more sun than on a bad day."

Point to: the card "The consequence"

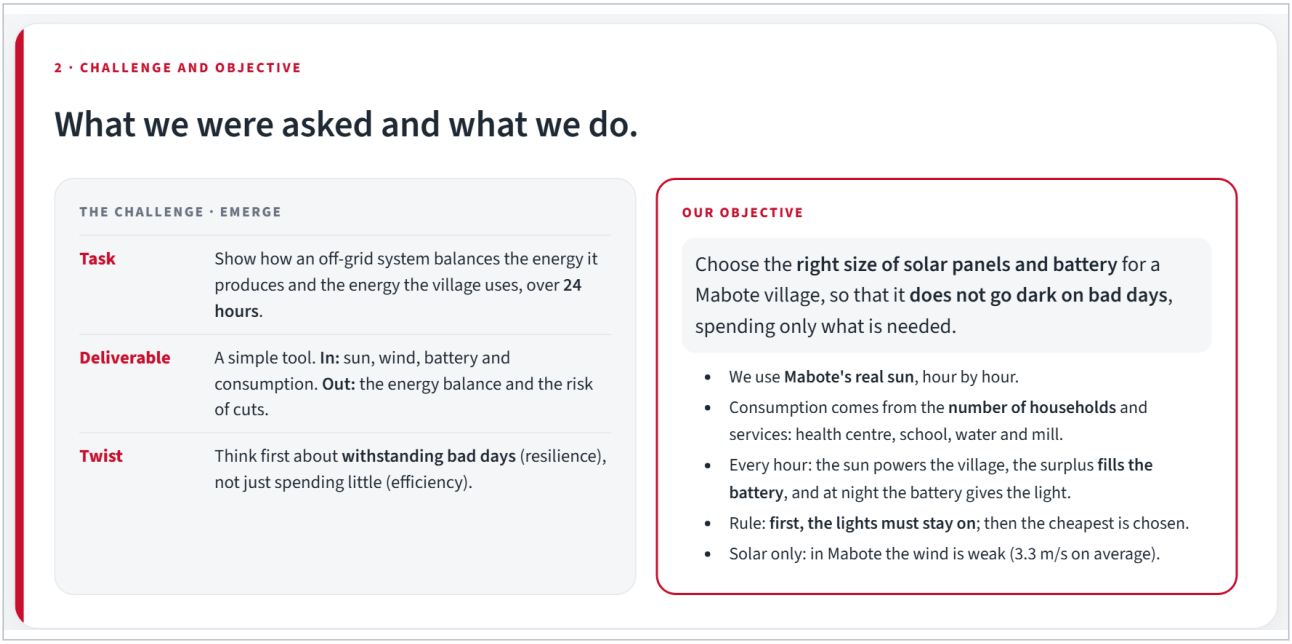
4. "If the system is designed only for the average day, on bad days the village, the health centre and the water pump lose power."

Key sentence: "The problem is not the average day. It is the bad days."

Hand-over to the next slide: *"This is what the EMERGE challenge asked us to solve."*

Slide 2 · Challenge and objective

Time: 60 s · **Speaker:** Laura · **Goal:** say what we were asked, our objective and how it works.



What to say

Point to: the grey box "The challenge"

1. "The challenge asked for three things: show how an off-grid system balances the energy it produces and the energy the village uses, over 24 hours; a simple tool that shows the energy balance and the risk of power cuts; and to think first about withstanding bad days, not just about spending little."

Point to: the red box "Our objective"

2. "Our objective is to choose the right size of solar panels and battery for a Mabote village, so that it does not go dark on bad days, spending only what is needed."

Point to: the points in the list

3. "We use Mabote's real sunshine, hour by hour, and the consumption of 200 households and services. Every hour, the sun powers the village, the surplus fills the battery, and at night the battery provides the light. The rule is simple: first, the lights must stay on; then we choose the cheapest option."

Point to: the last point

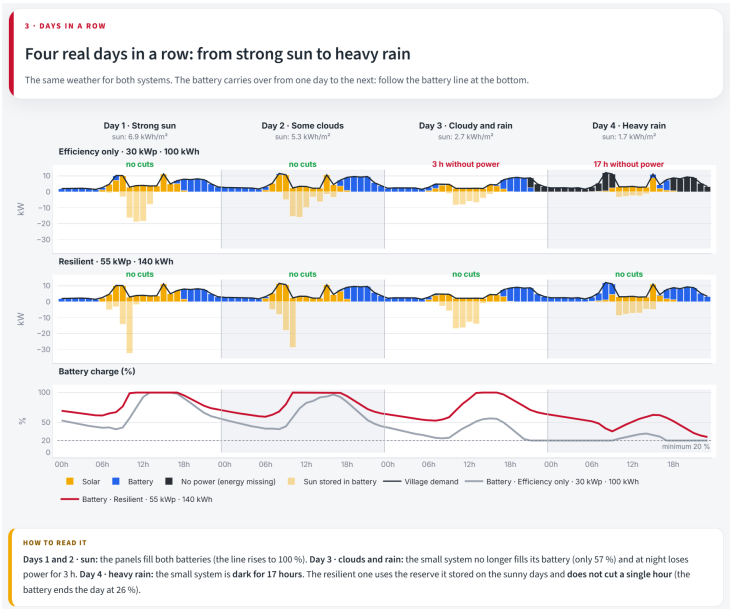
4. "And why solar only? In Mabote the wind is weak: 3.3 metres per second on average."

Key sentence: "First we make sure the lights stay on; then we choose the cheapest."

Hand-over to the next slide: *"I will hand over to Suely, who will show this working over four real days in a row."*

Slide 3 · Days in a row

Time: 100 s · **Speaker:** Suely · **Goal:** the jury sees when the battery fills, when it runs out and why one system fails.



What to say

Point to: the day names, at the top

1. “These are four real consecutive days in Mabote, from strong sun to heavy rain. Both systems get exactly the same weather. At the top, the cheapest system, with 30 kWp of panels and 100 kWh of battery. In the middle, our resilient system, with 55 kWp and 140 kWh.”

Point to: the colours

2. “Yellow is energy from the sun, blue is energy from the battery, and black is when the village has no power.”

Point to: the battery line, at the bottom

3. “The most important thing is this line: how much the battery holds. The grey line is the cheap system and the red line is ours.”

Point to: days 1 and 2

4. “On the sunny days, the panels fill both batteries to 100 %. Nobody loses power.”

Point to: day 3

5. “On the cloudy, rainy day, the cheap system can no longer fill its battery: it only reaches 57 %. At night the battery runs out and the village is without power for 3 hours. Ours still fills its battery.”

Point to: day 4

6. “On the heavy-rain day, the cheap system leaves the village in the dark for 17 hours: that is the black. Ours uses the energy it stored on the sunny days and does not cut a single hour.”

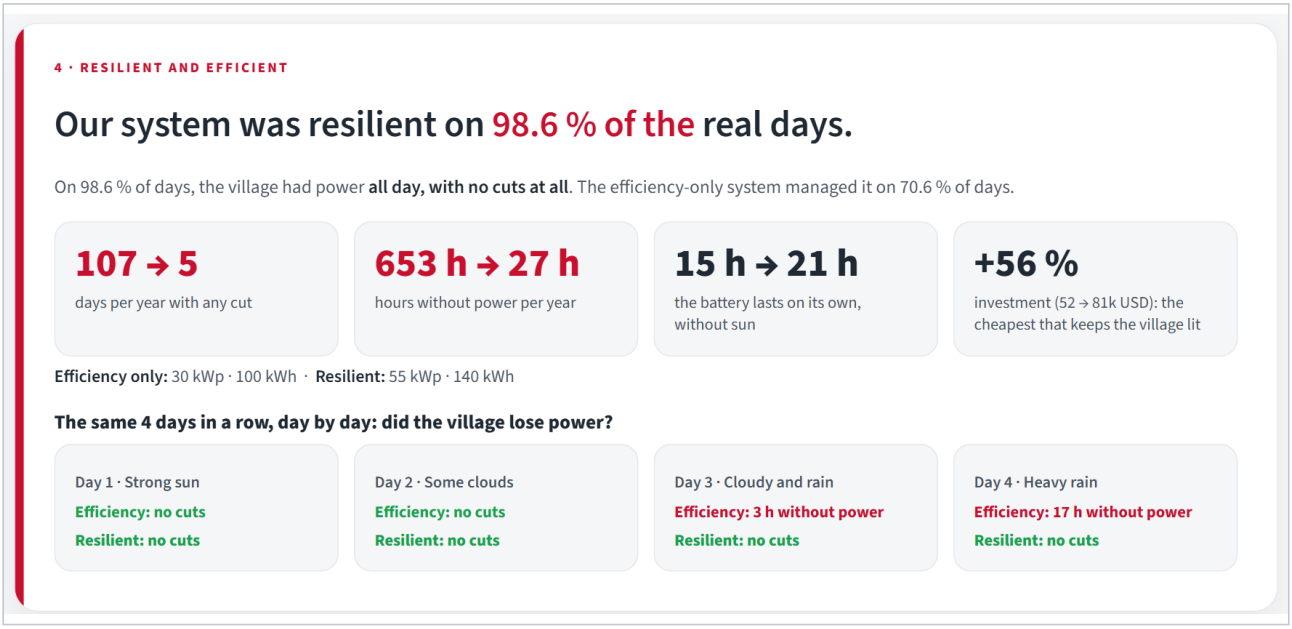
Tip: always point to the battery line. It tells the story: it rises with the sun and falls at night. Pause after “17 hours in the dark”.

Key sentence: “Sunny days store energy for rainy days. But only if the system is the right size.”

Hand-over to the next slide: *“That was a four-day example. Now Ezequias will show the result over all the real days.”*

Slide 4 · Resilient and efficient

Time: 90 s · **Speaker:** Ezequias · **Goal:** state, in plain words, how resilient the system was.



What to say

Point to: the title

1. "Our system was resilient on 98.6 % of the real days. That means: on almost every day, the village had power all day, with no cuts at all. The cheap system only managed this on 70.6 % of days."

Point to: the big cards

2. "In number of days: the cheap system has cuts on 107 days a year; ours on 5. Hours without power fall from 653 to 27 a year. And the battery can run the village on its own for 21 hours without sun."

Point to: the "+56 %" card

3. "This costs 56 % more investment. But it is the cheapest system that does not leave the village in the dark. That is why it is both resilient and efficient."

Point to: the 4 cards at the bottom

4. "At the bottom, the same four days as the previous slide: the cheap system cuts 3 hours on day 3 and 17 on day 4. Ours: no cuts on any of the four."

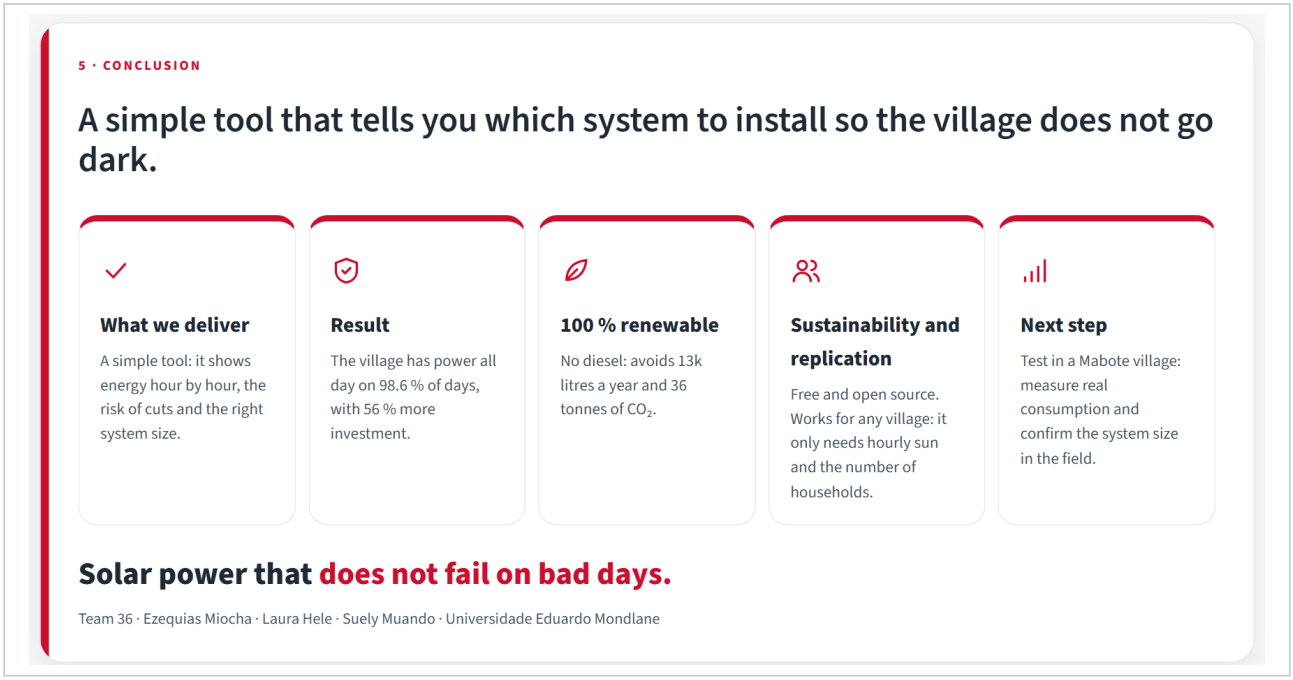
Tip: say the number in the title slowly: "98.6 percent of days". It is the sentence the jury should remember.

Key sentence: "Our system was resilient on 98.6 % of days."

Hand-over to the next slide: *"To finish, what this means for the village and for the future."*

Slide 5 · Conclusion

Time: 60 s · **Speaker:** Ezequias · **Goal:** close with the value, the clean energy and the next step.



What to say

Point to: the card "What we deliver"

1. "We deliver a simple tool: it shows energy hour by hour, the risk of power cuts and the right system size for the village."

Point to: "Result" and "100 % renewable"

2. "The village has power all day on 98.6 % of days. And it is all solar energy: no diesel, avoiding about 13 thousand litres a year and 36 tonnes of CO_2."

Point to: "Sustainability" and "Next step"

3. "The tool is free and works for any village: it only needs hourly sunshine and the number of households. The next step is to test it in a Mabote village, measure real consumption and confirm the system size."

Key sentence: "Solar power that does not fail on bad days. Thank you."

Hand-over to the next slide: *"(End. Thank the jury and invite questions.)"*

2. Live demo (optional)

If the jury asks to see the platform working, or during the questions. About 1 minute.

Step	Where	What to do	What to say
1	24-h simulator	Choose **Day 4 · Heavy rain** and press **Efficiency only** in the left panel.	“This is the cheap system on a rainy day: look at the black.”
2	24-h simulator	Press **Resilient**.	“With the resilient system, the black disappears.”
3	Left panel	Raise the **Number of households** from 200 to 300.	“If the village grows, demand rises and cuts appear again: the tool shows it straight away.”
4	System sizing	Show the recommended system and press **Use this system in the simulator**.	“The platform recalculates the right system for the new village.”
5	Solar energy	Show the green cards.	“100 % renewable: diesel and CO_2 avoided, and the cost of energy.”

Golden rule for the demo: first say what is going to happen, then press the button. If something takes time to load, keep talking with the PDF or the slide.

3. Questions and answers

The jury's most likely questions, with short answers. Answer calmly, in two or three sentences, and if you do not know something, say it is a point for the pilot.

Essential questions (the jury's criteria)

What is the innovation of the project?

The innovation is not the panel or the battery: it is **how the system is chosen**. (1) We size the system for **bad days in a row**, testing thousands of real days in the order they happened, instead of the average day or a “typical year”, which hides the extremes. (2) **Resilience is required** and only then is the cheapest option chosen; the usual practice is the opposite. (3) The result is told in the village's language: “resilient on 98.6 % of days”, “hours without power”. (4) It shows that resilience is possible with **100 % renewable energy**, without a diesel generator. If asked about HOMER: it exists and is good, but it is paid software, needs specialists and usually uses a typical year.

How replicable is it?

Highly. The tool needs only two things: the **hourly sunshine** of the site and the **number of households** and services. Sunshine data come from ERA5, free for any point in the world: you only change the coordinates in the data script. No weather station or prior measurements are needed. The code is open and the platform runs online or on an ordinary computer.

Is it scalable?

Yes. Sizing one village takes seconds (156 systems tested over thousands of days in about 3 seconds). For a whole district, all you need is a list of villages with their coordinates and number of households. It works for villages of 50 to 500 households, with or without a health centre, school, water pump and mill.

What is the local impact?

The village has power all day on 98.6 % of days, and the health centre, water pump and school no longer stop on rainy days. It avoids about 13 thousand litres of diesel and 36 tonnes of CO_2 a year. And the solar surplus can power new productive uses during the day: cooling, milling, irrigation.

Is it feasible?

Yes. It uses mature technology sold in Mozambique: about 100 panels of 550 W (about 260 m²) and LFP lithium batteries, with an investment of about 81 thousand USD. The tool already works and is online.

How is it financially sustainable?

The tool is free and open source, maintained by the team at UEM. The village system can be financed by rural electrification programmes and partners. The diesel avoided alone is worth about 19 thousand USD a year: the investment is recovered in about 4 years of fuel saved (fuel only, not counting the generator or its maintenance).

Why Mabote?

It is a rural district of Inhambane where the grid does not reach most villages. It has good sunshine (5.4 kWh/m² a day on average), but also cloudy and rainy days in a row. It represents many villages in southern Mozambique.

Can the team take this forward?

We are three UEM students who built the model, the data processing and the platform from scratch, and it is already online. For the pilot, we are looking for technical support from EMERGE and local partners.

About the challenge

The challenge mentions wind energy. Why don't you use it?

We analysed Mabote's real wind: the average is 3.3 m/s, too weak for a small wind turbine, which needs stronger wind to produce well. That is why the system is solar with batteries only, which is cheaper and more reliable here. Wind is included in the model because it cools the panels, which slightly increases output. At a windier site, the model could include a wind turbine.

What is the difference between efficient and resilient?

Efficient means not spending or wasting more than needed. Resilient means continuing to supply power on bad days: clouds, heavy rain, heat. Our system is the cheapest among the resilient ones: it is both.

Isn't the resilient system just a bigger battery?

No. A battery does not produce energy; it only stores what the panels produce. With few panels, on a cloudy day the battery does not even fill up. Reaching the same resilience with battery alone would need 500 kWh, about 180 thousand USD, more than twice our system. The resilient system combines more panels (30 to 55 kWp) with a slightly larger battery (100 to 140 kWh).

How do you measure the risk of power cuts?

In three simple ways: the energy missing (what the village asked for and did not get, in %), the hours without power, and bad days per year, which are days with more than 10 % of the energy missing.

About the data and the model

What does “resilient on 98.6 % of days” mean?

That on 98.6 % of the real days tested, the village had power for all 24 hours, with no cuts. On the rest, about 5 days a year, there was some cut, almost always short and during bad days in a row. The efficiency-only system has power all day on only 70.6 % of days.

Why show four days in a row instead of a single day?

Because the battery carries over from one day to the next. What matters is not one isolated bad day, but what happens when bad days come after others: if the battery did not fill the day before, the village goes dark. The four days show this directly: sunny days store energy for rainy days.

What is the methodology based on?

Standard photovoltaic engineering formulas: radiation on the panel plane (isotropic sky model), output with 12 % system losses and the temperature effect (−0.35 % per °C, cell temperature by the NOCT method, with wind cooling), and a battery model with 95 % efficiency and a 20 % minimum charge. Everything is explained, with the formulas, on the platform's Methodology page (it is not in the presentation, to save time).

Where do the data come from?

From the ERA5 reanalysis of the European Copernicus programme (ECMWF): real sunshine, temperature and wind in Mabote, hour by hour. We used 4,018 real days, about 11 years of data.

Why test so many days? Isn't one day enough?

One day shows how the system works, and that is what the 24-hour simulator does. But to choose the system we need to know how often bad days come and how many come in a row. In Mabote up to 5 weak days can come in a row. If we looked at a single year, it could be a lucky year.

Why does the battery start the day with different charges?

Because energy carries over from one day to the next. Sunny days fill the battery for the following days; after rainy days, the battery arrives emptier. The simulator starts each day with the charge left by the previous real days.

How do you calculate demand?

From the number of households (about 0.3 kWh per household per day: lights, phone, radio, fan) and the services: health centre, school, water pump, mill and street lighting. For 200 households this is about 122 kWh a day. It varies with the hour, the day of the week, the heat and from day to day.

Why does the battery never go below 20 %?

To protect the life of the lithium (LFP) battery. Full discharges greatly shorten battery life.

Has the model been validated?

It uses standard solar engineering formulas (output with the temperature effect, 95 % battery efficiency) and internationally validated reanalysis data. Demand is a typical estimate: the next step is a pilot to measure it in the field and fine-tune the model.

How do you choose the recommended system?

We test 156 combinations of panels and batteries. The resilient system is the cheapest that meets three rules: at most 0.5 % of energy missing, at most 3 bad days a year and at least 12 hours of autonomy.

About results and costs

How much does the system cost?

The resilient system costs about 81 thousand USD (55 kWp of panels and 140 kWh of batteries), compared with 52 thousand USD for the efficiency-only system. These are indicative African mini-grid costs: 650 USD per kWp and 320 USD per kWh.

Isn't +56 % too expensive?

It is the cheapest system that does not leave the village in the dark. And the energy costs 0.17 USD/kWh over 20 years, including maintenance and battery replacement, while the fuel alone for a diesel generator costs 0.43 USD/kWh.

The resilient system has 51 % surplus. Isn't that waste?

It is the price of resilience: to fill the battery on a rainy day, there is energy left over on sunny days. But this surplus is an opportunity: it can power new uses during the day, such as vaccine cooling, milling, irrigation or phone charging. It is available for the village to grow.

Why not add a backup diesel generator?

Diesel is expensive, hard to transport to remote villages and polluting. We show that it is possible to be resilient with 100 % renewable energy, avoiding about 13 thousand litres of diesel and 36 tonnes of CO_2 a year.

How long can the battery last without sun?

About 21 hours on its own in the resilient system, compared with 15 hours in the efficiency-only system.

About impact and the future

What if the village grows?

There is a demand-growth control in the panel. The platform immediately shows the cuts that appear, and the sizing page recalculates the right system.

What about a cyclone?

The system is tested on real heavy-rain days, which is what happens to energy during a storm. Physical protection of the equipment (structures and fixings) is a separate engineering step, outside the scope of this tool.

Who will use the tool?

People who plan mini-grids: rural electrification technicians, NGOs, local authorities and communities. It is simple enough to use without special training.

What is the next step?

A pilot in a Mabote village: measure real demand for a few months, compare it with the model and confirm the system size before installing.

> **If you don't know the answer:** “That's a good question. Our model does not cover that yet, and it is one of the points we want to confirm in the pilot.” Never make up numbers.

4. Numbers to memorise

Reference village: 200 households, with a health centre, school, water pump and mill.

Number	What it means
30 kWp · 100 kWh	“Efficiency only” system: the cheapest for the average day
55 kWp · 140 kWh	Recommended resilient system (100 panels of 550 W, about 260 m²)
52 → 81 thousand USD (+56 %)	Investment of the two systems
70.6 % → 98.6 %	Resilient days: the village had power all day, no cuts
107 → 5	Days per year with any cut
62 → 2.7	Bad days per year (more than 10 % of energy missing)
17 h → 0 h	Hours without power on day 4 (heavy rain) of the 4-day sequence
653 h → 27 h	Hours without power per year
15 h → 21 h	Battery autonomy without sun
180 thousand USD	Cost of reaching the same resilience with battery only (30 kWp + 500 kWh)
122 kWh/day	Average village demand (ranges from 93 to 149)
5.4 kWh/m²/day	Average sunshine in Mabote (ranges from 2.3 to 8.2)
13 days/yr · 5 in a row	Days with less than half the normal sunshine; longest run
3.3 m/s	Average wind: too weak for a wind turbine
156	Systems tested in the sizing
100 %	Renewable energy (zero fuel)
13.4 thousand litres/yr	Diesel avoided (about 19 thousand USD a year)
36 t/yr	CO_2 avoided (716 t over 20 years)
0.17 vs 0.43 USD/kWh	Cost of solar energy over 20 years vs generator fuel alone
24 % · 26 % · 51 %	Solar energy: straight to the village · stored in the battery · free surplus

Sentences to never forget

- “Our system was resilient on 98.6 % of days.”
- “The problem is not the average day. It is the bad days.”
- “First we make sure the lights stay on; then we choose the cheapest.”
- “The cheap system works on good days. Ours works every day.”